



Argand Diagrams · Mark Scheme

Core Pure 1 Chapter 2: Argand Diagrams. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Modulus and argument

- 1 $|z| = 5$ B1; $\arg z = -\arctan \frac{4}{3} = -0.927$ B1.
- 2 z_1 at $(2, 1)$ and z_2 at $(-1, 3)$ B1; $z_1 + z_2 = 1 + 4i$ at $(1, 4)$ B1.
- 3 Moduli 2 and $\sqrt{2}$ B1; $\arg(-\sqrt{3} + i) = \frac{5\pi}{6}$ B1; $\arg(-1 - i) = -\frac{3\pi}{4}$ B1.
- 4 $z = \frac{(1+7i)(1+i)}{2}$ M1; $= -3 + 4i$ A1; $|z| = 5$ B1; $\arg z = \pi - \arctan \frac{4}{3} = 2.21$ B1.
- 5(a) $|z| = \sqrt{4+12} = 4$ B1; $\arg z = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ B1.
- 5(b) $4 - 8\sqrt{3}i + 12i^2$ M1; $= -8 - 8\sqrt{3}i$ A1.
- 5(c) $|z^2| = 16$ B1; $\arg(z^2) = -\frac{2\pi}{3}$ (not $\frac{4\pi}{3}$) B1.
- 5(d) z in the second quadrant at $(-2, 2\sqrt{3})$ B1; z^2 in the third quadrant at $(-8, -8\sqrt{3})$ B1.

Section B – Modulus–argument form

- 6 $4\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$ M1; $= 2 + 2\sqrt{3}i$ A1.
- 7 $r = \sqrt{2}$ B1; $\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$ B1.
- 8 $r = \sqrt{3+9} = 2\sqrt{3}$ B1; third quadrant: $\theta = -\pi + \arctan \sqrt{3}$ M1; $2\sqrt{3}\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right)$ A1.
- 9 Multiply moduli, add arguments M1; $z_1 z_2 = 6\left(\cos \frac{5\pi}{12} + i\sin \frac{5\pi}{12}\right)$ A1; divide moduli, subtract arguments M1; $\frac{z_1}{z_2} = \frac{2}{3}\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$ A1.
- 10(a) $1 + \sqrt{3}i = 2\left(\cos \frac{\pi}{3} + i\sin \frac{\pi}{3}\right)$ B1; $1 - i = \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$ B1.
- 10(b) $r = \frac{2}{\sqrt{2}} = \sqrt{2}$, $\theta = \frac{\pi}{3} - \left(-\frac{\pi}{4}\right)$ M1; $z = \sqrt{2}\left(\cos \frac{7\pi}{12} + i\sin \frac{7\pi}{12}\right)$ A1.
- 10(c) $z = \frac{(1+\sqrt{3}i)(1+i)}{2}$ M1; $= \frac{1-\sqrt{3}}{2} + \frac{1+\sqrt{3}}{2}i$ A1; real parts: $\sqrt{2}\cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{2}$, so $\cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{2}-\sqrt{6}}{4}$ A1.

Section C – Loci

- 11 A circle with centre $(3, -2)$, i.e. $3 - 2i$ B1, and radius 4 B1.
- 12 Half-line starting at $(1, 0)$, with the point itself excluded B1; at angle $\frac{\pi}{4}$ to the positive real direction, into the first quadrant B1.
- 13 $(x-4)^2 + y^2 = x^2 + (y+2)^2$ M1; $-8x + 16 = 4y + 4$ M1; $y = 3 - 2x$ A1.
- 14 (a) $(x+1)^2 + (y-2)^2 = 9$ B1. (b) Distance from O to the centre $(-1, 2)$ is $\sqrt{5}$ M1; maximum $3 + \sqrt{5}$ A1; minimum $3 - \sqrt{5}$ A1.
- 15(a) Circle centre $(0, 3)$, radius 3, touching the real axis at O B1; half-line from $(-3, 0)$ (excluded) at $\frac{\pi}{4}$, passing through the centre of the circle B1.
- 15(b) $y = x + 3$ B1, $x > -3$ B1.
- 15(c) $x^2 + (x+3-3)^2 = 9$ M1; $x = \pm \frac{3\sqrt{2}}{2}$, both > -3 A1; $z = \frac{3\sqrt{2}}{2} + \left(3 + \frac{3\sqrt{2}}{2}\right)i$ and $z = -\frac{3\sqrt{2}}{2} + \left(3 - \frac{3\sqrt{2}}{2}\right)i$ A1.



Section D – Regions and combined loci

16 Circle centre $(2, 0)$, radius 2, through O $B1$; inside shaded (boundary included) $B1$.

17 (a) $|z - 3 - 2i| \leq 2$ $B1$; $\operatorname{Re}(z) \geq 3$ $B1$. (b) Semicircle: $\frac{1}{2}\pi(2)^2 = 2\pi$ $B1$.

18 $(x-6)^2 + y^2 \leq x^2 + (y-2)^2 \Rightarrow -12x + 36 \leq -4y + 4$ $M1$; $4y \leq 12x - 32$, i.e. $y \leq 3x - 8$ $A1$; line through $(\frac{8}{3}, 0)$ and $(4, 4)$, region below/right of it shaded $B1$.

19 Circle centre O radius 4 $B1$; half-lines from O at $\frac{\pi}{6}$ and $\frac{2\pi}{3}$ $B1$; sector between them shaded $B1$; angle $\frac{\pi}{2}$, area $\frac{1}{2}(16)(\frac{\pi}{2}) = 4\pi$ $B1$.

20(a) $(x-4)^2 + y^2 = x^2 + (y-4)^2$ $M1$; $-8x = -8y$, so $y = x$ $A1$.

20(b) Circle centre $(4, 2)$, radius 2, touching the real axis $B1$; line $y = x$ cutting the circle $B1$; the part of the disc below the line $y = x$ (containing the centre) shaded $B1$.

20(c) Distance from $(4, 2)$ to $y = x$ is $\frac{|4-2|}{\sqrt{2}} = \sqrt{2}$ $M1$; $\cos \frac{\alpha}{2} = \frac{\sqrt{2}}{2}$, so the chord subtends $\alpha = \frac{\pi}{2}$ at the centre $A1$; minor segment $= \frac{1}{2}(2)^2(\frac{\pi}{2} - \sin \frac{\pi}{2}) = \pi - 2$ $M1$; area of $R = 4\pi - (\pi - 2) = 3\pi + 2$ $A1$.

Roar-level challenge

21 Extreme arguments occur where the half-line from O is tangent to the circle, giving a right-angled triangle with hypotenuse 4 and opposite side 2 $M1$; angle from the imaginary axis $= \arcsin \frac{1}{2} = \frac{\pi}{6}$, so greatest $\arg z = \frac{2\pi}{3}$ $A1$; least $\arg z = \frac{\pi}{3}$ $A1$; at the tangent point $|z| = \sqrt{16-4} = 2\sqrt{3}$ $M1$; $z = 2\sqrt{3}(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}) = -\sqrt{3} + 3i$ $A1$.